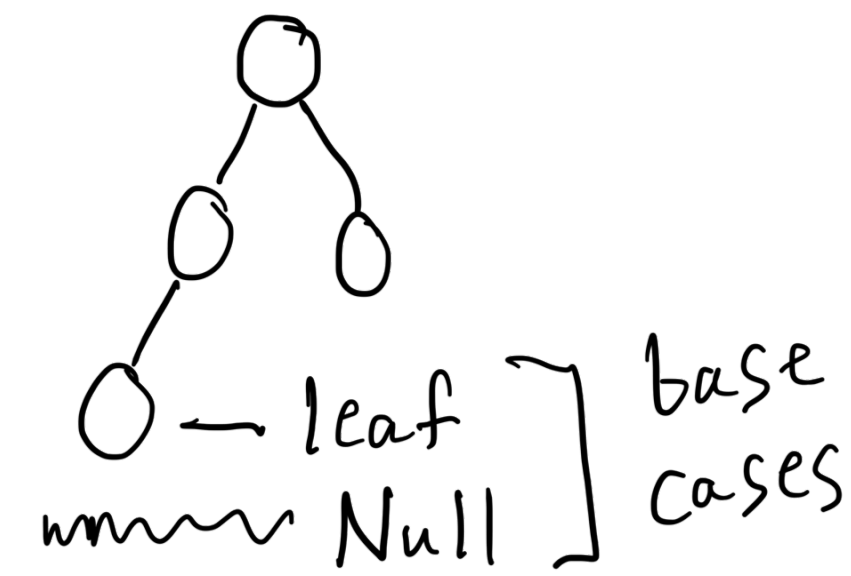


12. Count leaves on tree

For the following question, assume binary trees consist of nodes from the following class:

```
class node {
public:
    double data;
    node * left;
    node * right;
    node (double x) {data = x; left = nullptr; right = nullptr;}
};

// Function to count the leaf nodes in a binary tree
int countLeaves(node* root) {}
```

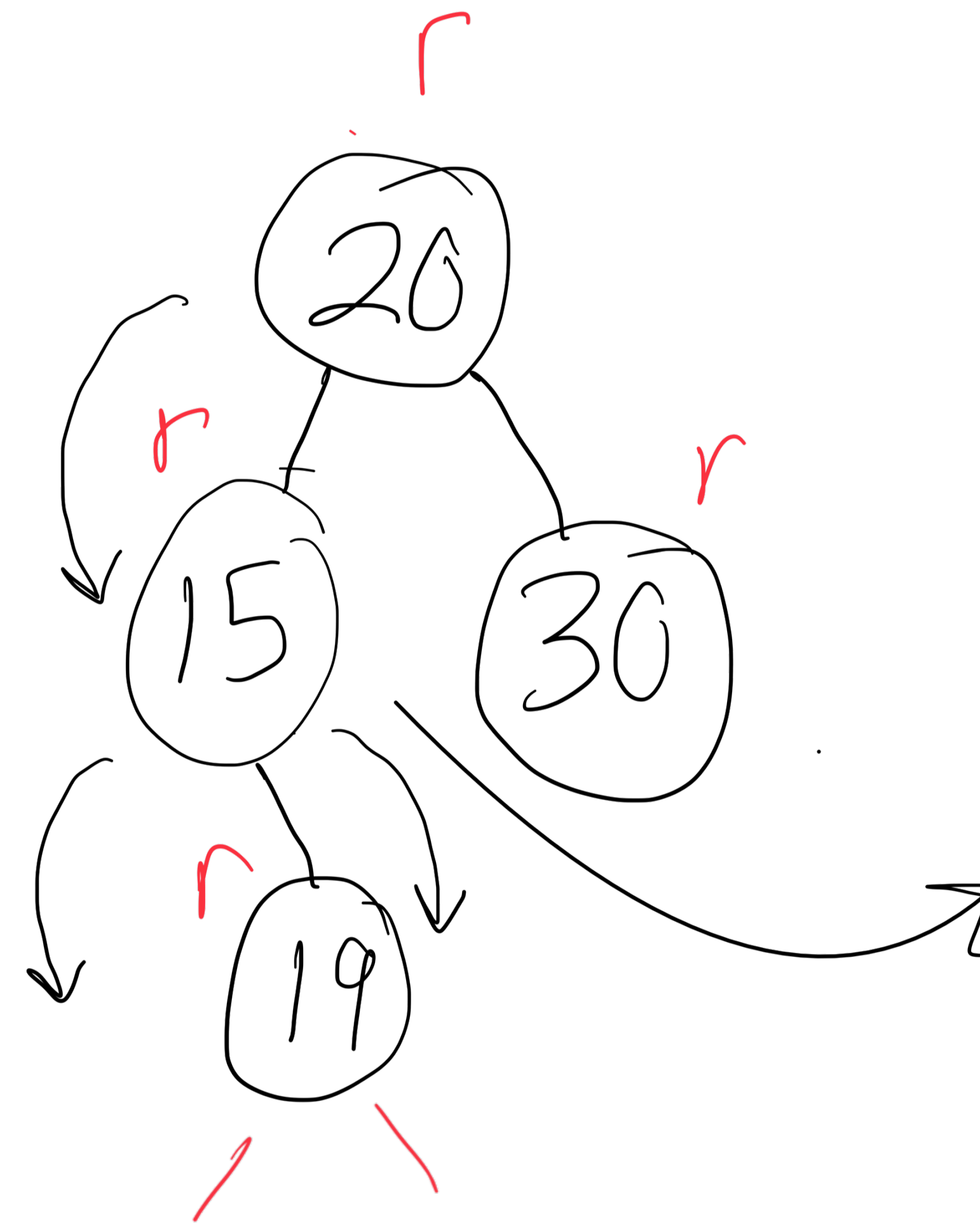


```
int countLeaves (node* root) {
    if (root == nullptr) {return 0; } X X ✓ X
    if (root → left == nullptr &&
        root → right == nullptr) X X ✓
    { return 1; }
    return countLeaves (root → left) +
           countLeaves (root → right);
}
```

3

root = 20
 root = 15
 root = null
 root = 19
 ⋮
 ⋮
 ⋮

return L recursive + R recursive = 1 + 1 = 2
 return L recursive + R recursive = 0 + 1 = 1
 return 0
 return 1



Computer says: Nice!
You have 1 leaf
Subtree 15

19
null
15
20

Recursion is
a stack